Today's Agenda

1) Grouped Query Attention (GQA)
2) MLA (DeepSeek)
3) Paged Attention (KV Cache Management)
4) ROPE
   - MLA

RECAP

MHA → 8 Heads $H_1 \to Q_1, K_1, V_1$; $H_2 \to Q_2, K_2, V_2$; ... $H_8 \to Q_8, K_8, V_8$

MQA → 8 Heads $H_1 :: Q_1, K_1, V_1$; $H_2 :: Q_2, K_1, V_1$; ... $H_8 : Q_8, K_1, V_1$

Reduce :- KV Heads

GQA → H_1 ... H_8 → one KV head shared

{ Attention Diversity } Losing

GQA → Middle ground (MHA - MQA)

Reduce the loss of info

Assumption

H_1 Q_1] → $K_1 V_1$
H_2 Q_2

H_3 Q_3] → $K_2 V_2$
H_4 Q_4

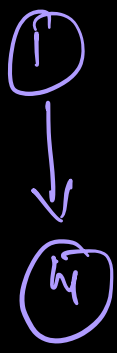
H_5 Q_5] → $K_3 V_3$
H_6 Q_6

H_7 Q_7] → $K_4 V_4$
H_8 Q_8

(K, V)

head → Hyperparameter.

KV (Head) = 4



512
64

MHA
64
64
64

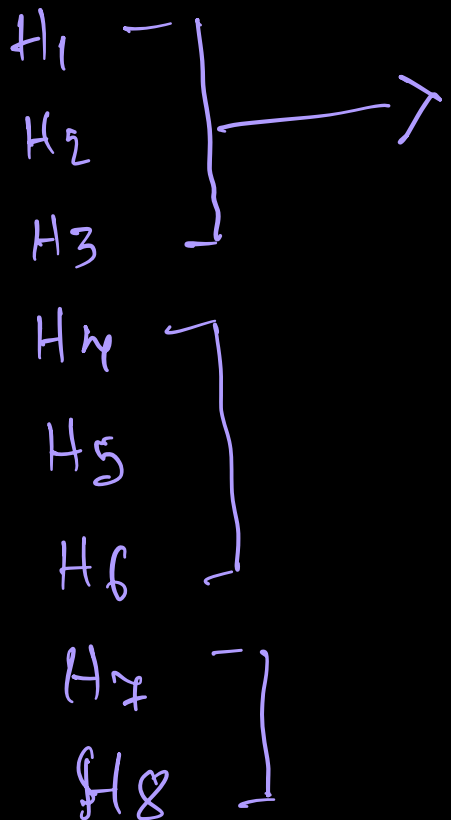
diff KV

MQA
64
KV 64
Same

GQA
64
$K_1 V_1$ = 64
64
$K_2 V_2$ = 64
64

MHA = 50% Reduction
50% Info loss

MQA → MHA
12.5 %



MHA :- 32 KV Heads - 100% KV Memory

GQA :- 16 KV Heads → 50% KV Memory

OQA :- 8 KV Heads → 25%

QQA :- 4 KV Heads → 12.5%

MQA :- 1 KV Head → 3.12%

$$H_KV = \frac{H_Q}{4}$$

(Middle ground) — Practical Design Rule

4Q Heads → 1 KV Head

27 / 3 = 9 (Whole No)

Design

Hidden Dimension = 4096

Query Heads = 32

Head Dimension = 128

MHA

32Q = 32K

$\frac{32\,Q}{4}$ = 8 KV heads

32Q = 1 KV

MQA

Inference Efficiency

$$\text{Sharing Ratio} = \frac{\text{Q Heads}}{\text{KV Heads}}$$

(4) → Gold Rule

A100 → 40 GB VRAM



→ OOM

Llama 3 13B

Idea

↓

Training

Inference

2) Paged Attention {VLLM}

↳ Smart KV cache management.

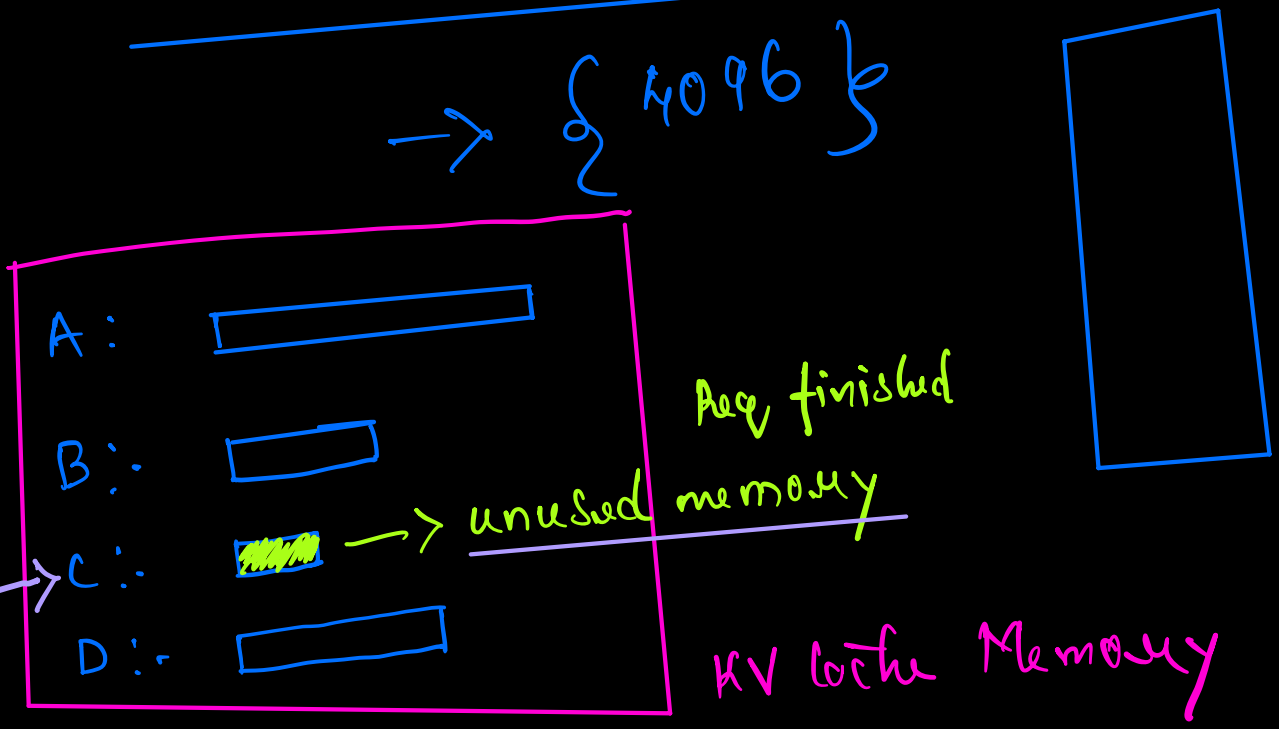
Problem :-

R_1	User A :-	2000 Tokens
R_2	User B :-	500 Tokens
R_3	User C :-	300 Tokens
R_4	User D :-	1000 Tokens

Allocation of Contigous Memory

Max Tokens Output/ Context Window

→ {4096}



F :- 2048

① Fragmentation

② Unknown Sequence length

No allocation with contiguous memory

Problems

KV cache = Small fixed size pages / blocks

Intuition

Page 0 :- Tokens 0 - 15

Page 1 :- Tokens 16 - 31

Page 2 :- Tokens 32 - 47

Page 3 :- Token 48 — 63

. . . .

Logical

Page 7 :: Token 0 - 15

Page 1 16 - 31

Page 13 32 - 47

Page 11 48 - 63

GPU Block Table

Block	
	0 → GPU Block 7
	1 → GPU Block 2
	2 → 15
	3 → 4

Attention

$K = [K_1, K_2 \ldots\ldots K_n]$

$V = [V_1, V_2 \ldots\ldots V_n]$

Block 7 :- $K_1 \ldots K_{16}$

2 :- $K_{17} \ldots K_{32}$

15 :- $K_{33} \ldots K_{48}$

4 :- $K_{49} \ldots K_{64}$

Kernel :- Block mapping to memory.

vLLM
Sglang

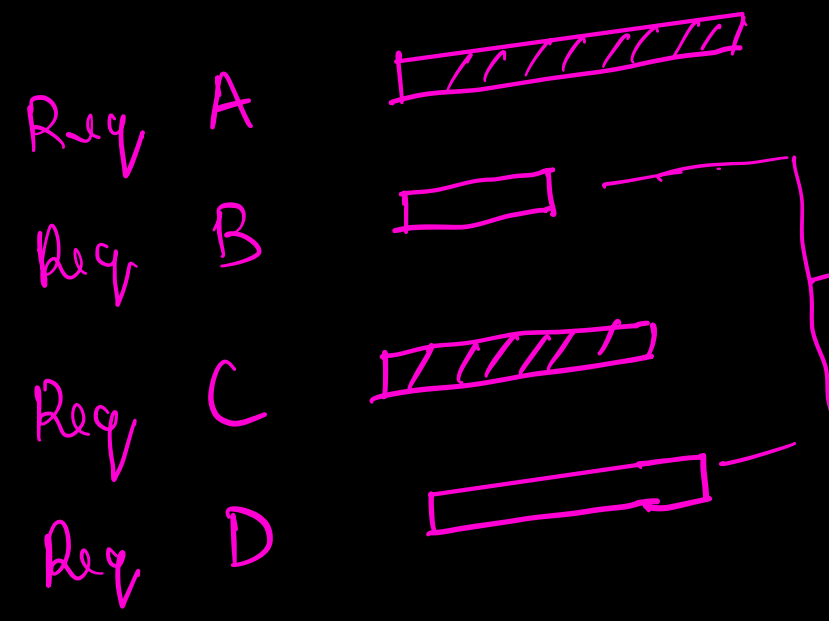
Fragmentation → Internal
→ External

Internal

I.F → Allocated memory is larger than what you need.

External

E.F → Free memory is there, but scattered into diff req.



New Req = more memory than the FREE Units.

10 units

Solve the Fragmentation

B 7 :- Token 0 - 15

B 13 :

B 1 :

B 21 : Token . . . .

Non- contigous Memory

External → Reduced

Without Paging



contigous

Paged Attention

Block 7

1

11

17

Scattered Memory → useful.

GPU Memory Region

Internal Frag

B : 16 Tokens

Seq : 18 Tokens

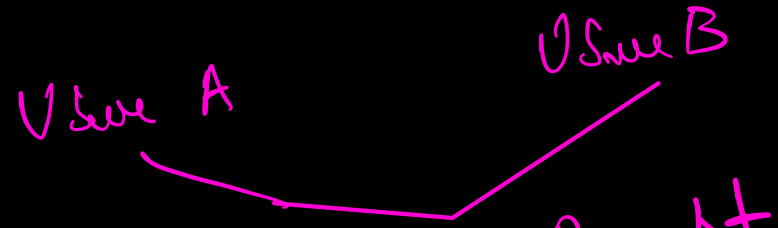
B1 : 16

B2 : 2 → 14 Token wasted

Only partial filled blocks Ly waste

One block / Sequence

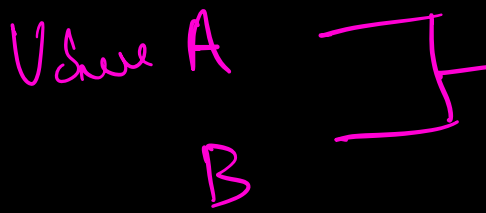
Duplications



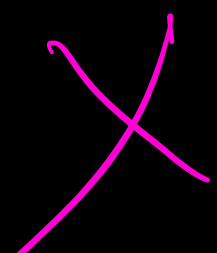
User A :- KV
B :- KV

Same info !!, Twice

Paged Attention → Block Sharing / Page



User A → Block 17
B → Block 25



One physical copy → Serves both requests.

Prefix sharing → KV sharing

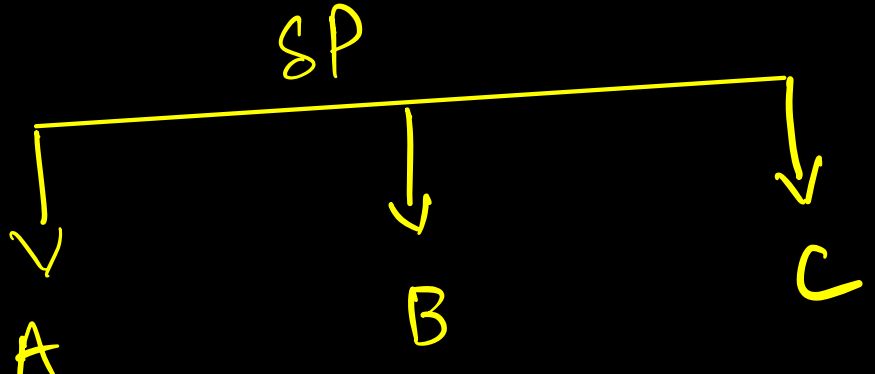
Chat Systems

- SYSTEM PROMPT

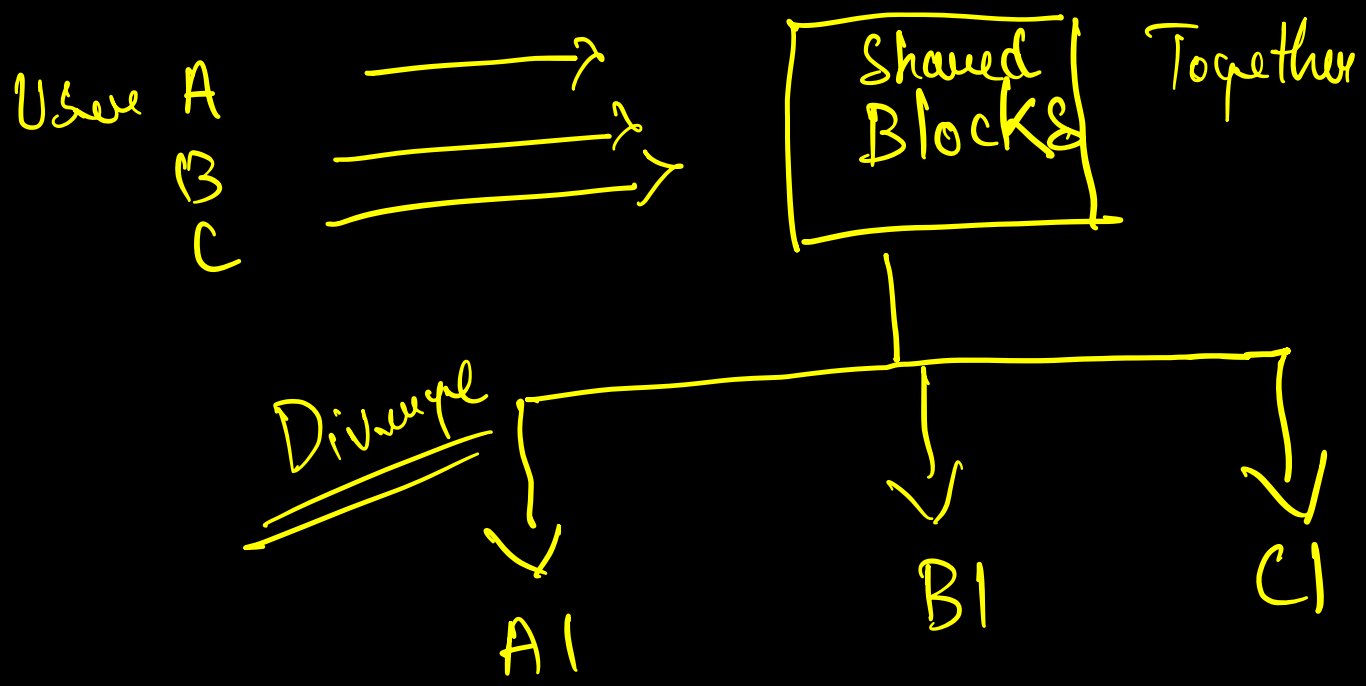
User A : SP + Prompt A

User B : SP + P B

C : SP + P C



(SP) Paged K/V will be used by other user



SP + Prompt A = Prompt B = Prompt

Prompt -> 100 Tokens

A - 50 Sony 50 diff

B - 50 Same ,

C - 50 Same ,

1) Block Mapping

Block level hash

hashing / prefix caching

Same Block

↓

Same Hash

↓

Same Prefix

Prefix Caching

KV-Cache Sharing

Reference Counting

User A :- Same P

User B :- Same Prompt

Reference count = 2

A

C = 1

B =

C = 0

MLA (DeepSeek)

Multi-head Latent Attention

MHA, MQA, GQA

KV

Idea :- No store K, V

Smaller latent Representation

Compression

↳ Store

$K = x W_K$

$V = x W_Q$

C = latent vector

Latent Dim = 128

x ↓ compression ↓ latent C ↓ store C

$K = 512$

$V = 512$

$KV = 1024$

MHA

GQA = $K = 4 \times 64 = 256$

$U = 25$

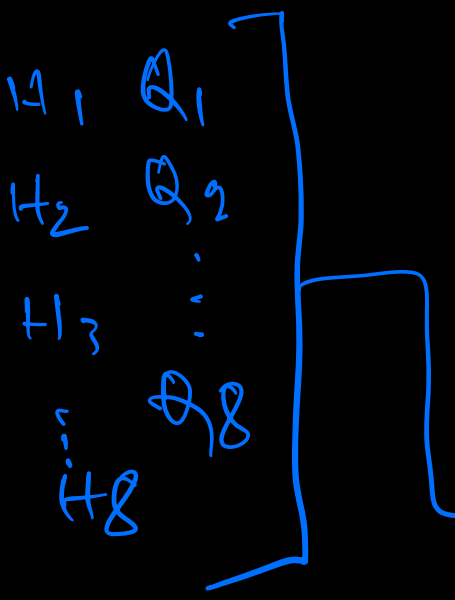
$KV = 512$

MQA = K = 64
V = 64
KV = 128

Assume

Latent Dimension = 64

(K, V) both



8 nos

Smaller matrix projection

W_{DKR}

Projection

3 nos

X

↓

compression

↓

C_KV ← cache

↳ Expanded

↓ K related representation

↓ V related representation

$C_{KV} = X \; W_{DKV}$

smaller matrix projection

$K = C_{KV} \; W_K$

$V = C_{KV} \; W_V$

K, V → compressed → original exact KV

old / Token 1 → K_1, V_1

2 → K_2, V_2

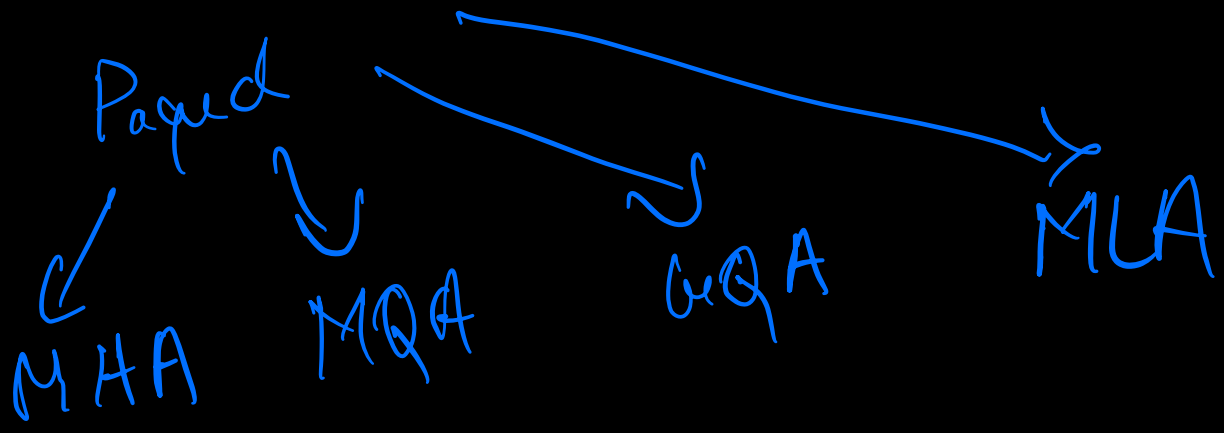
Token → C_1

Token → C_2

→ C_3

X P.E → ROPE

TE + PE = X


Paged
MHA
MQA
GQA
MLA